Language Development 1: Speech Perception

Nature and Nurture in Child Language

"Nativist" positions: Remember Chomsky's critique of Skinner

The important feature of language is its *generativity*.

Acquiring a language = identifying the abstract grammar that produces sentences in the language.

General learning mechanisms studied by behaviorists were not up to this task:

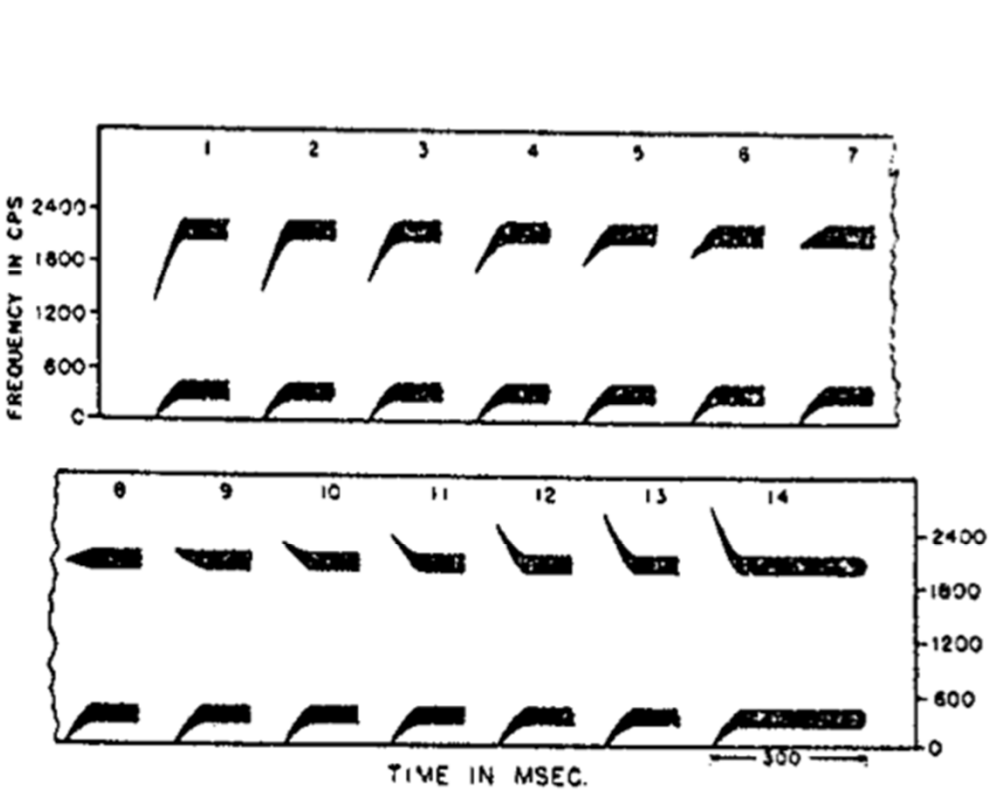
- grammatical and ungrammatical sentences are not "similar" in any straightforward way.
- The context for language acquisition does not look like operant conditioning – reward/punishment contingencies don't hold

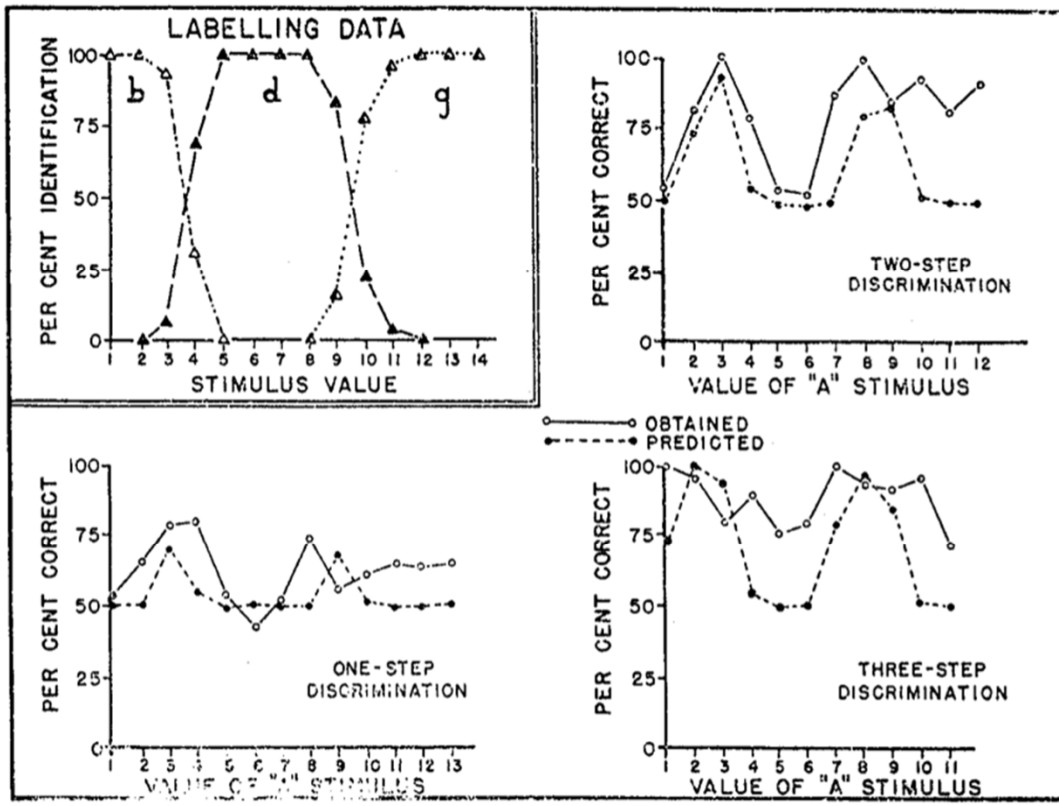
Instead, nativist approaches assert:

- innate *knowledge* of what language is like (Universal Grammar)
- language-specific acquisition processes

"The fact that all normal children acquire essentially comparable grammars of great complexity with remarkable rapidity suggests that human beings are somehow specially designed to do this, with data-handling or 'hypothesis-formulating' ability of unknown character and complexity."

Remember Categorical Perception





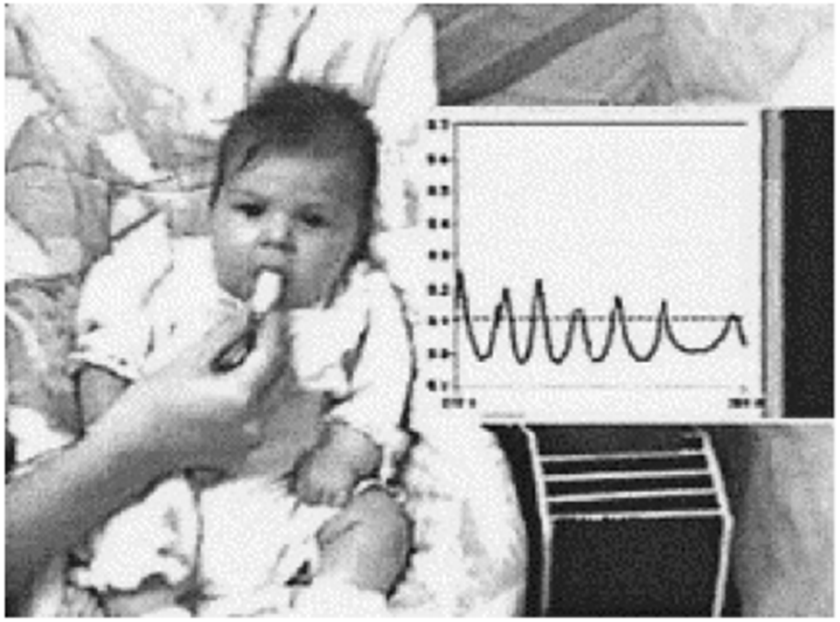
Early theories of categorical perception treated it as a *language-specific* and *species-specific* phenomenon. Only humans were known to show categorical perception, and only for speech sounds.

It's a compelling example of how similarity between stimuli is defined in *linguistic*, rather than *physical* terms.

Categorical perception in infants



Infant in the "conditioned head turn procedure from Janet Werker's website:
http://infantstudies.psych.ubc.ca/



We'll also look at "high amplitude sucking." (image from an unrelated paper, Rochat & Striano, 2001, IBD)

Categorical perception in infants

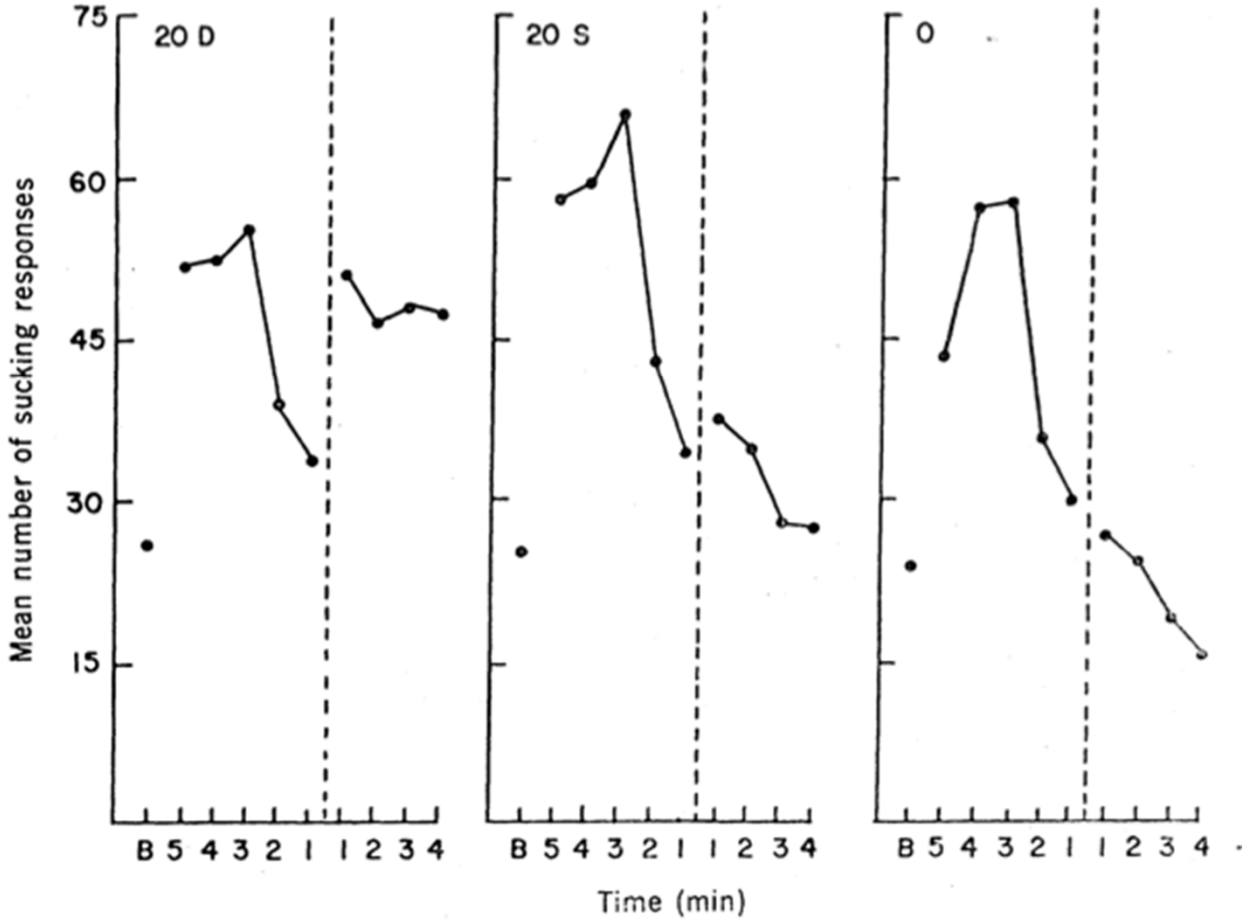
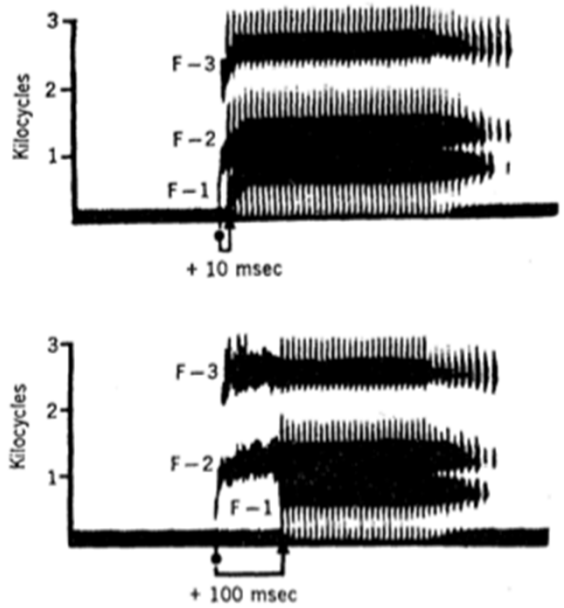


Fig. 2. Mean number of sucking responses for the 4-month-old infants, as a function of time and experimental condition. The dashed line indicates the occurrence of the stimulus shift, or in the case of the control group the time at which the shift would have occurred. The letter *B* stands for the baseline rate. Time is measured with reference to the moment of stimulus shift and indicates the 5 minutes prior to and the 4 minutes after shift.

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In the main experiment, there were three variations in VOT differences at each of two age levels. In the first condition, 20D, the difference in VOT between the two stimuli to be discriminated was 20 msec and the two stimuli were from different adult phonemic categories. The two stimuli used in condition 20D had VOT values of +20 and +40 msec. In the second condition, 20S, the VOT difference was again 20 msec, but now the two stimuli were from the same phonemic category. In this condition the stimuli had VOT values of −20 and 0 msec or +60 and +80 msec. The third condition, 0, was a control condition in which each sub-

Eimas et al. (1971, Science)

Why was this a big deal?

Categorical perception was considered to be the first step in turning speech into a *symbolic* format that could be used by later linguistic processes.

Four month olds are really little. Some gross motor milestones for that age include:

- manipulating objects with two hands
- holding their head and chest up while prone



Their color vision and general visual acuity are still “in progress.”

They are just starting to produce vowel-like sounds instead of creaky whines.

This is very, very early in development for an apparently language-specific ability to emerge, so it seemed like evidence for innateness.

“Perceptual narrowing” for speech categories

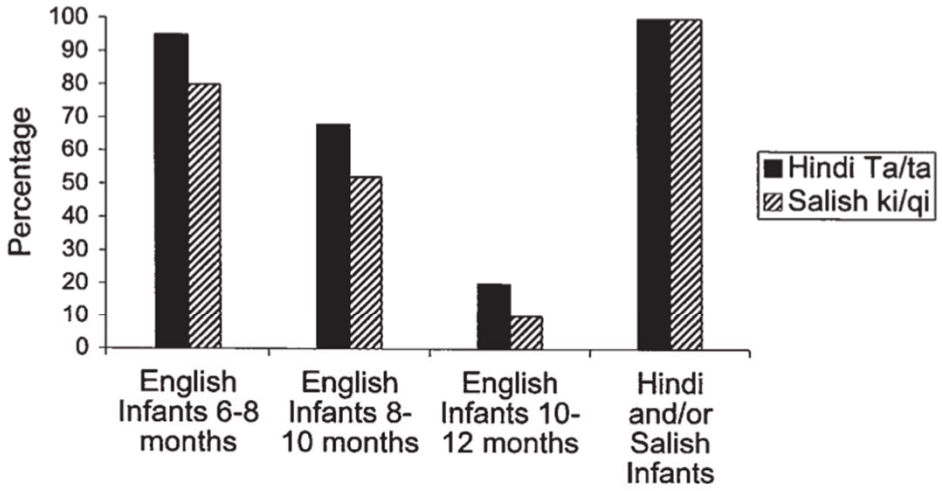


FIGURE 3 The effects of age and experience on infants' ability to discriminate two non-English contrasts. The proportion of infants at each age reaching discrimination criterion on the Hindi and Salish (non-English) contrasts. From "Cross-language speech perception: Evidence for perceptual reorganization during the first year of life," by J. F. Werker & R. C. Tees, 1984a, *Infant Behavior and Development*, 7, 49–63. Adapted with permission of the authors.

At around six months, English-learning infants show evidence of discriminating speech contrasts they've never heard before.

Over the course of development, the ability to perceive these contrasts declines.

How is this related to categorical perception?

A very compelling narrative:

Speech perception depends on categorical perception abilities that are a special property of the language system.

These are present very early in development.

In fact, children start off as "citizens of the world," able to hear all contrasts (i.e., they have *innate, language-general knowledge about speech*) but then rapidly tune their perceptual systems to the specific language in their environment.

The role of *learning*, then, is to *select* among the many possible innately specified phonetic contrasts, and *narrow* down the inventory to just those that are meaningful in the child's native language.

Some problems for this view

Categorical perception:

- not limited to language
- not limited to humans
- task-dependent

Infant speech perception:

- not actually “universal”

Categorical perception of non-speech sounds

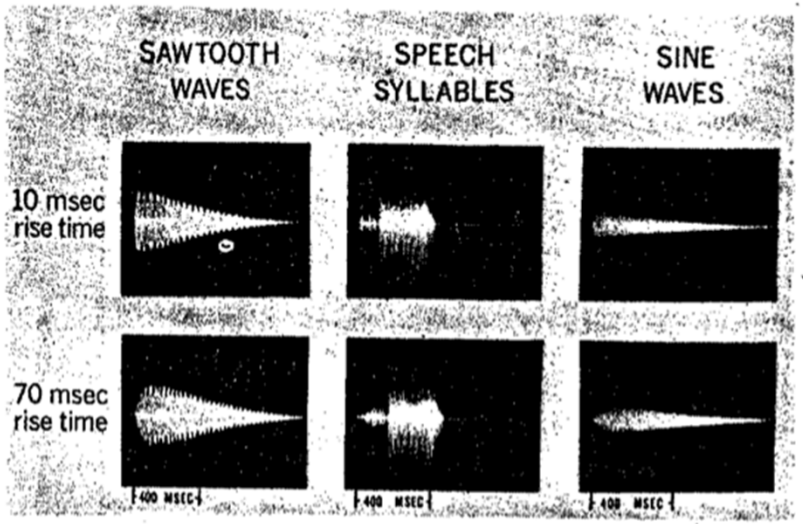
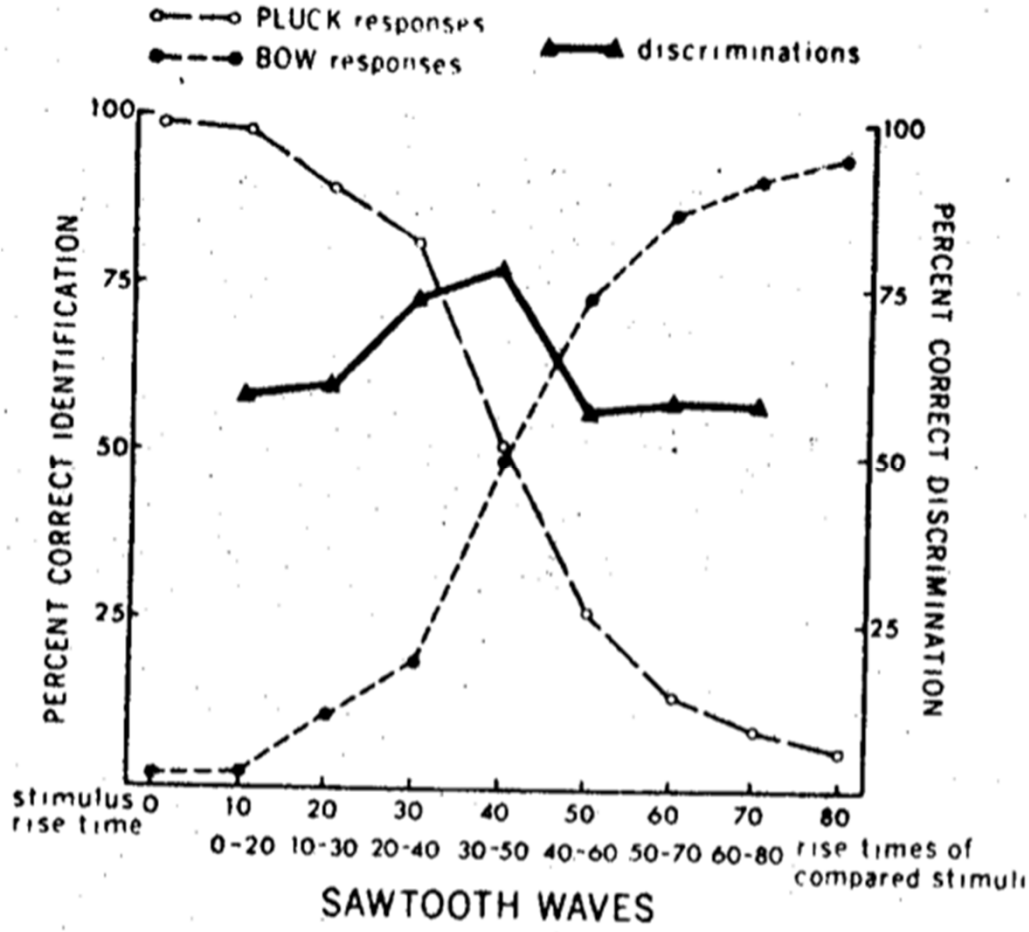


Fig. 1. Oscillograms of sample stimuli used in Experiments I and II.



Cutting & Rosner (1974, Perception & Psychophysics)

Categorical perception of non-speech sounds…by infants!

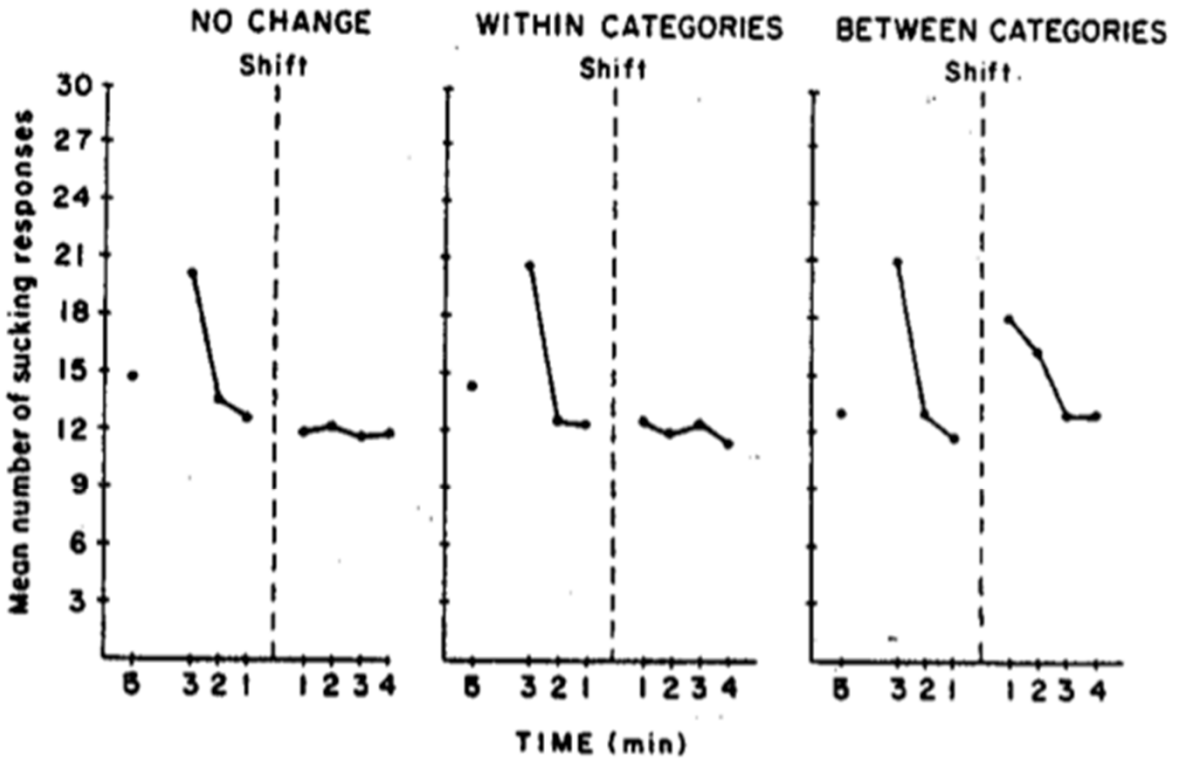


Figure 1. Mean sucking rates as a function of time and experimental session. Time is measured with reference to the moment of the stimulus shift, marked by the vertical dashed line. The baseline rate of sucking is indicated by the letter "B."

Jusczyk et al. (1977, Perception & Psychophysics)

Categorical perception of *speech* sounds…by chinchillas!



Trained using operant conditioning that either the 0-lag or 80ms-lag stimulus is a S^D for food.

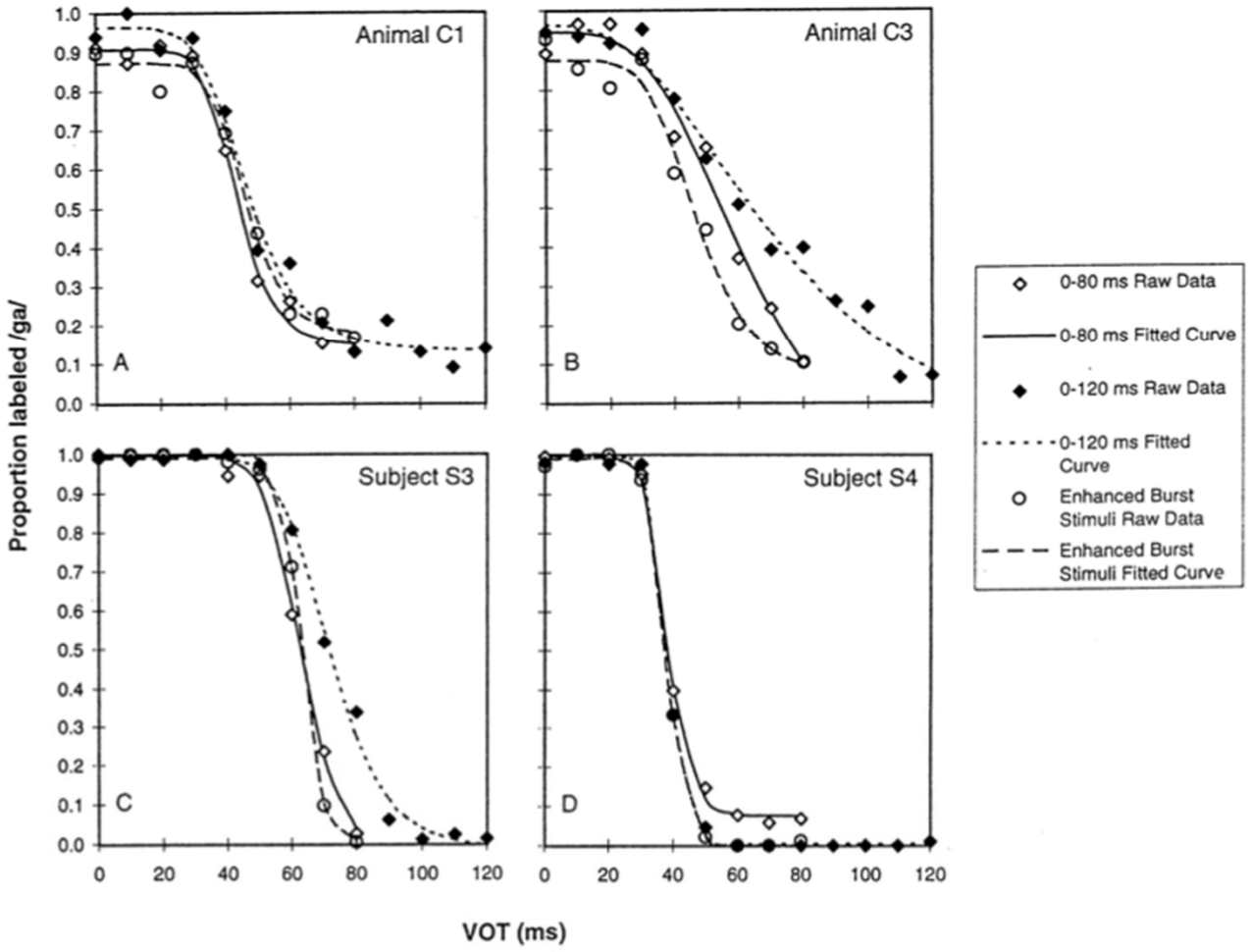


Fig. 4. Raw data and fitted curves for two chinchilla and human listeners for the /ga/–/ka/ continua. Increasing the intensity of the frication burst and aspiration noise (enhanced burst stimuli) shifted the category boundary to a shorter VOT and/or increased the slope of the curve one animal (B) and one human listener (C).

Ohlemiller et al.(1999, Behavioral Brain Research, see also Kuhl & Miller, 1975)

Some contrasts are less salient than others for infants.

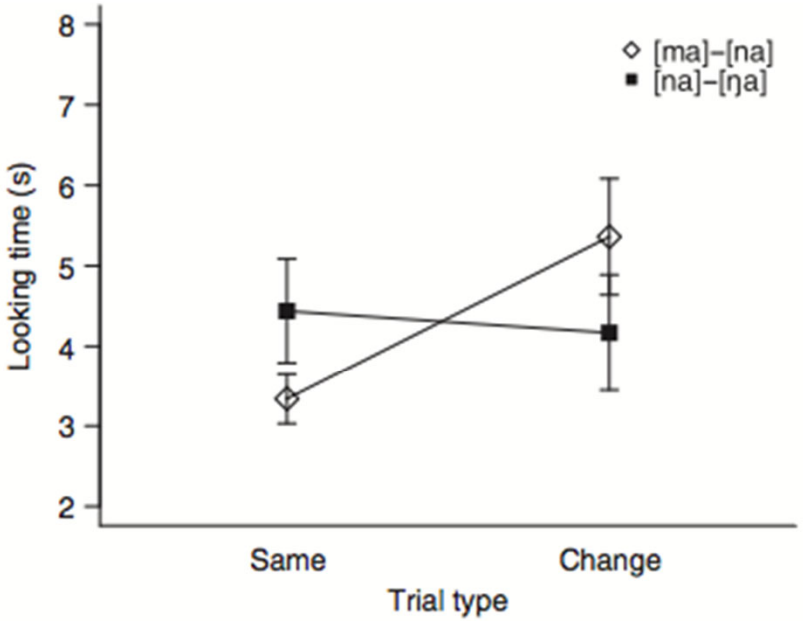


Figure 4 *English-hearing infants' (at 4–5 months) looking time to same vs. change trials for the native [ma]–[na] and non-native [na]–[ŋa] contrasts. Error bars represent standard error.*

A place contrast for nasals in onset (/na/-/ŋa/) is not readily perceived by four month olds, independent of language experience.

In contrast /ma/-/na/ is relatively salient

(Narayan et al., 2010)

Some contrasts are less salient than others for infants.

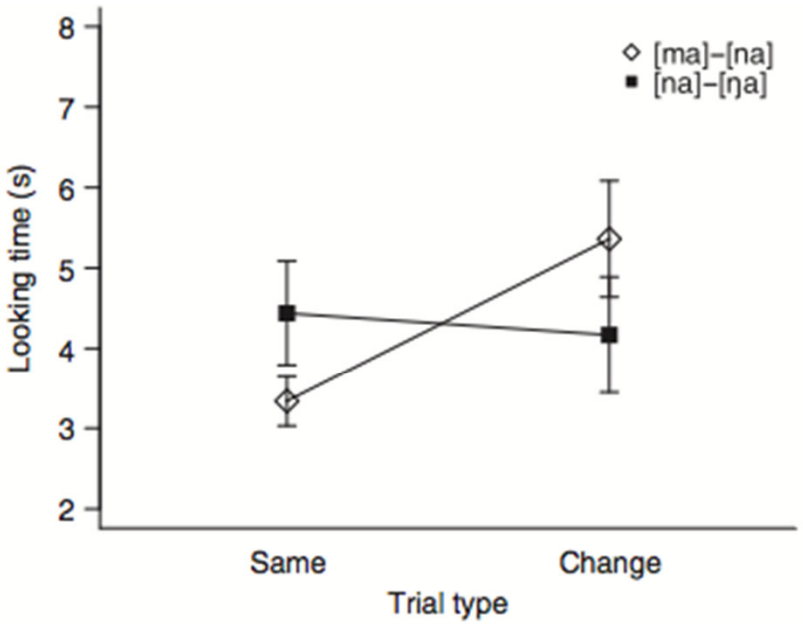


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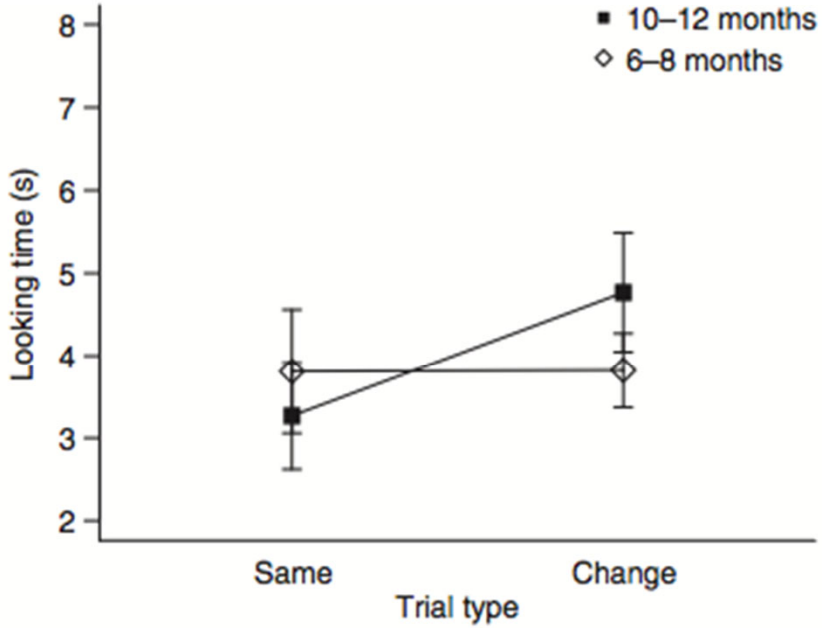


Figure 5 *Filipino-learning infants' (at 6–8 and 10–12 months) looking time to same vs. change trials for the native [na]–[ŋa]. Error bars represent standard error.*

Between 6 and 10 months, exposure to this contrast in a Filipino language environment increases its salience.

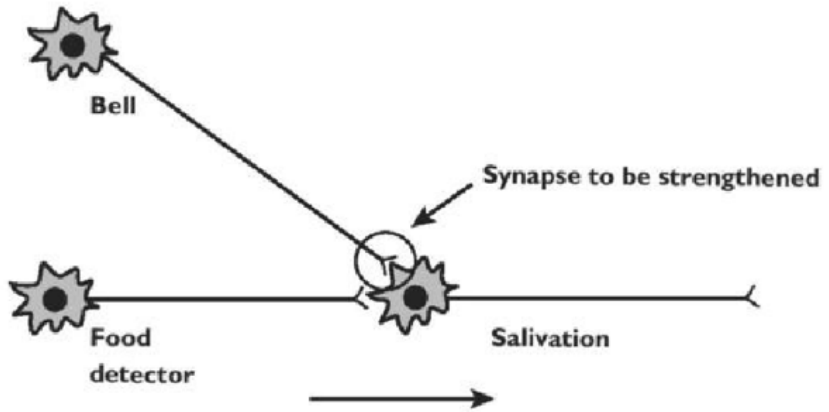
So, it can't be that all speech contrasts are available innately. Exposure to contrasts in the environment is necessary for *acquisition* in some cases.

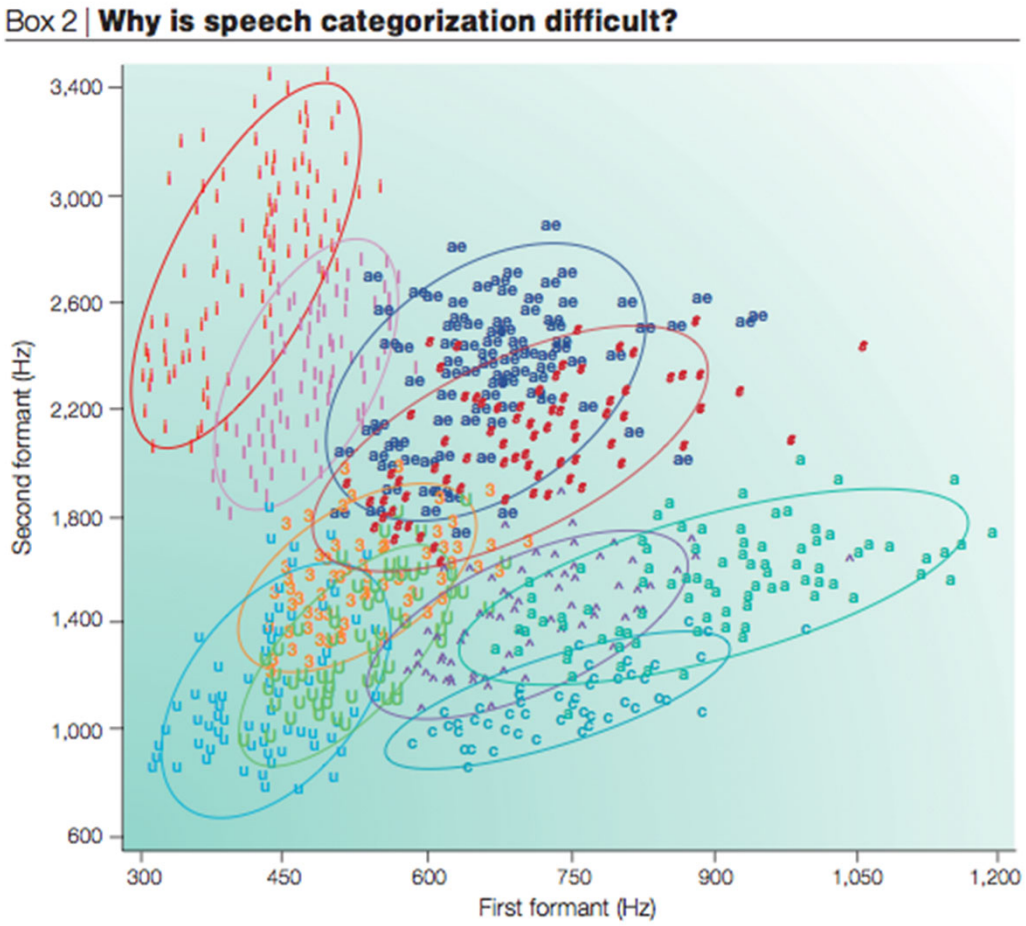
(Narayan et al., 2010)

OK, so what is the alternative to the nativist approach?

Emergentist approaches assume that associative learning mechanisms are sensitive to statistical regularities in the environment.

How can this can extend beyond *conditioning* to support learning for *pattern recognition?*





Here are the American vowels again!

When Kuhl and others talk about learning the statistics of distributions in the environment, they mean things like this.

We can think of a “vowel space” determined by the first and second formants.

Maybe we learn the vowels of our language by extracting the statistics of these distributions.

So, a “prototypical” /a/ is just the mean (or mode?) of the distribution of all the /a/ sounds we hear.

Does anything like that actually happen?

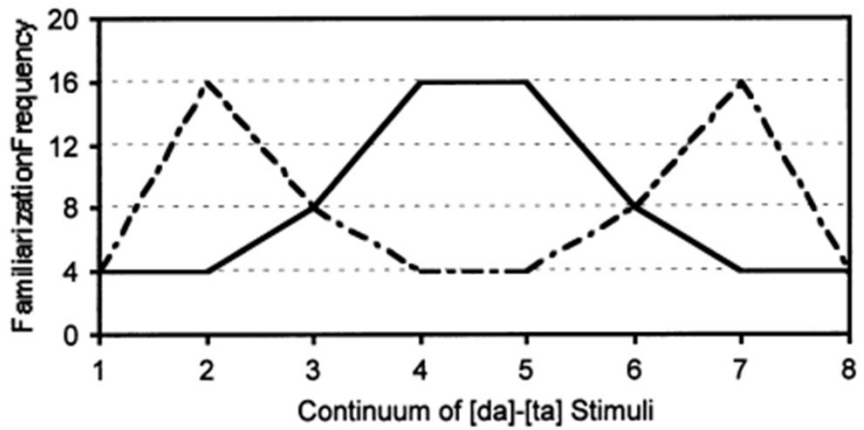


Fig. 1. Bimodal vs. Unimodal distributions of [da]–[ta] stimuli during familiarization. The continuum of speech sounds is shown on the abscissa, with Token 1 corresponding to the endpoint [da] stimulus, and Token 8 the endpoint [ta] stimulus. The ordinate axis plots the number of times each stimulus occurred during the familiarization phase. The presentation frequency for infants in the Bimodal group is shown by the dotted line, and for the Unimodal group by the solid line.

Table 1
Mean (SE) looking times for infants in each age group and familiarization condition on Alternating and Non-Alternating trials

	Alternating trials (s)	Non-Alternating trials (s)
6 months Unimodal	4.85 (0.47)	4.53 (0.51)
8 months Unimodal	4.98 (0.63)	5.20 (0.56)
6 months Bimodal	5.66 (0.44)	6.41 (0.32)
8 months Bimodal	5.45 (0.52)	6.15 (0.56)

Maye et al. had infants listen passively to two distributions of voice onset time.

The dashed lines indicate a *bimodal* distribution, where the input provided evidence for two different categories.

The solid line indicates a *unimodal* distribution, providing evidence for a single category.

At test, infants listened to stimuli 3 and 6 on the continuum in either alternating or non-alternating trials.

Only infants pre-exposed to the bimodal distribution were sensitive to the difference between conditions.

…as if the infants in the unimodal condition had learned to treat stimuli 3 and 6 as members of the same category.

Social/interactive context for learning.

Some experimental evidence that social interaction is important for learning:

The same exposure to a Mandarin speaker is more effective for learning when it happens in person (and is interactive) than when it is viewed as a video.

Social interactions contingent on production of different sounds seem to function much like a reinforcing stimulus.

One idea is that humans find social interaction with other humans innately *reinforcing*.

Another is that we are innately good at using *social cues* to learn from other people.

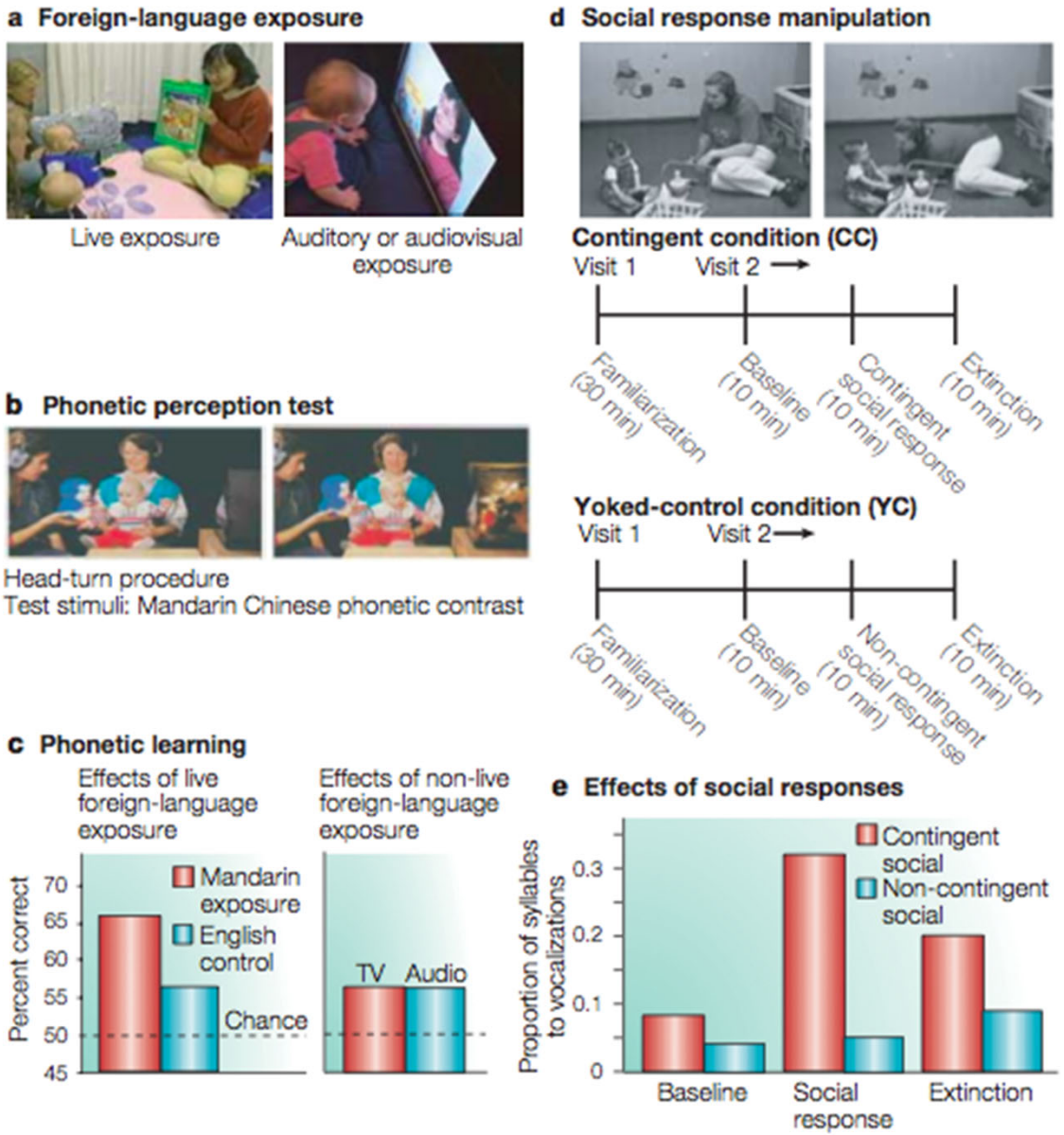


Figure 4 | **Two speech experiments on social learning. a** | Nine-month-old American infants being exposed to Mandarin Chinese in twelve 25-min live or televised sessions. **b** | After exposure, infants in the Mandarin exposure groups and those in the English control groups were tested on a Mandarin phonetic contrast using the head-turn technique. **c** | The results show phonetic learning in the live-exposure group, but no learning in the TV- or audio-only groups[31]. **d** | Eight-month-old infants received either contingent or non-contingent social feedback from their mothers in response to their vocalizations. **e** | Contingent social feedback increased the quantity and complexity of infants' vocalizations[80]. Panel **c** modified, with permission, from REF. 31 © (2003) National Academy of Sciences USA. Panels **d** and **e** modified, with permission, from REF. 80 © (2003) National Academy of Sciences USA.

Statistical learning with constraints

What do we mean by “constraints?” A lot of things:

1. What stimulus properties can be contrastive (*not* speaker, speech rate, etc.)

2. Constraints based on earlier learning (more about this when we discuss sensitive periods)

3. Social and/or interactive contexts for learning…